**JavaScript Day 2 Assignment**

**MAHIDDAR REDDY M**

**Associate Software Engineer**

Topics Covered: Printing Statements, Console Methods, Data Types, Operators, Arrays, Objects

Instructions: No loops. No conditional statements. Only use topics covered in class.

------------------------------------------------------------------------------------------------------------------------

1) Welcome Program

- Ask user name using prompt().

- Show alert: Hi , Welcome to JavaScript Training.

- Print the same message in console.

A) let userName = prompt("Please enter your name:"); // prompt for user to enter his name

let welcomeMessage = "Hi " + userName + ", Welcome to Js Training."; // creating welcome msg

alert(welcomeMessage); // Displaying the msg in alert box

console.log(welcomeMessage); // printing the msg to bowser console

2) Console Methods Practice

- Print 500 using console.log(), console.warn(), console.error().

- Use console.clear().

A) console.log(500); // Normal Output

console.warn(500); // Warning Msg

console.error(500); // Error Msg

console.clear(); // It clears the console above it

3) Data Type Identification

- Create string, number, boolean, undefined, and null variables.

- Print value and type using typeof.

A) let userName = "Mahiddar";   // String

let age = 21;  // Number

let isStudent = true;  // Boolean

let city; // Undefined

let score = null; // Null

console.log(userName, typeof userName);

console.log(age, typeof age);

console.log(isStudent, typeof isStudent);

console.log(city, typeof city);

console.log(score, typeof score); // null returns Object

4) Arithmetic Operations

- Use let a = 20 and let b = 5.

- Perform +, -, *, /, %, ** and print results.

A) let a = 20;

let b = 5;

// Arithmetic Operations

console.log("Addition:", a + b);

console.log("Subtraction:", a - b);

console.log("Multiplication:", a * b);

console.log("Division:", a / b);

console.log("Modulus (%):", a % b);

console.log("Exponential (**):", a ** b);

5) Increment & Decrement

- Demonstrate pre-increment, post-increment, pre-decrement, post-decrement.

- Print variables clearly.

A)// Initial value

console.log("Initial Value:", x);  // Output : 10

// Post-Increment

console.log("Post-Increment (x++):", x++); // Output : 10

// Uses current value, then increments

console.log("Value After Post-Increment:", x); // Output : 11

// Pre-Increment

console.log("Pre-Increment (++x):", ++x);  // Output : 12

// Increments first, then uses the value

// Post-Decrement

console.log("Post-Decrement (x--):", x--);  // Output : 12

// Uses current value, then decrements

console.log("Value After Post-Decrement:", x); // Output : 11

// Pre-Decrement

console.log("Pre-Decrement (--x):", --x);  // Output : 10

// Decrements first, then uses the value

6) Assignment Operators

- Use let num = 10.

- Perform +=, -=, *=, /=, %= and print results.

A) let num = 10;

num += 5;

console.log("After += 5:", num); // Output= 15 num -= 2;

console.log("After -= 2:", num); // Output= 13

num *= 3;

console.log("After *= 3:", num); // Output= 39

num /= 2;

console.log("After /= 2:", num); // Output= 19.5

num %= 4;

console.log("After %= 4:", num); // Output= 3.5

7) Array Access

- Create array ['HTML','CSS','JavaScript','React'].

- Print first element, second element, last element (using length), and total elements.

A) let technologies = ['HTML', 'CSS', 'JavaScript', 'React'];

// Accessing array elements

console.log("First Element:", technologies[0]);

console.log("Second Element:", technologies[1]);

console.log("Last Element:", technologies[technologies.length - 1]);

// Total number of elements

console.log("Total Elements:", technologies.length);

8) Modify Array

- Add one element at end.

- Remove last element.

- Print updated array.

A) let fruits = ['Apple', 'Banana'];

// Add one element at the end

fruits.push('Mango');

console.log("After Adding Element:", fruits);

// Remove the last element

fruits.pop();

console.log("After Removing Last Element:", fruits);

9) Student Object

- Create object with name, age, course, city.

- Print values using dot notation.

A) let student = {

   name: "Mahiddar Reddy",

   age: 21,

   course: "JavaScript Training",

   city: "Ongole"

};

console.log("Student Name:", student.name);

console.log("Age:", student.age);

console.log("Course:", student.course);

console.log("City:", student.city);

10) Nested Object Practice

- Create company object with nested trainer object.

- Print company name, trainer name, trainer subject.

A) let company = {

   companyName: "Stackly",

```
  trainer: {
    trainerName: "Naveen Kumar",
    subject: "JavaScript & Web Development"
  }
};
console.log("Company Name:", company.companyName);
console.log("Trainer Name:", company.trainer.trainerName);
console.log("Trainer Subject:", company.trainer.subject);
```